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(54) **VOLLEY TRAINING APPARATUS**

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ABSTRACT

Publication Classification

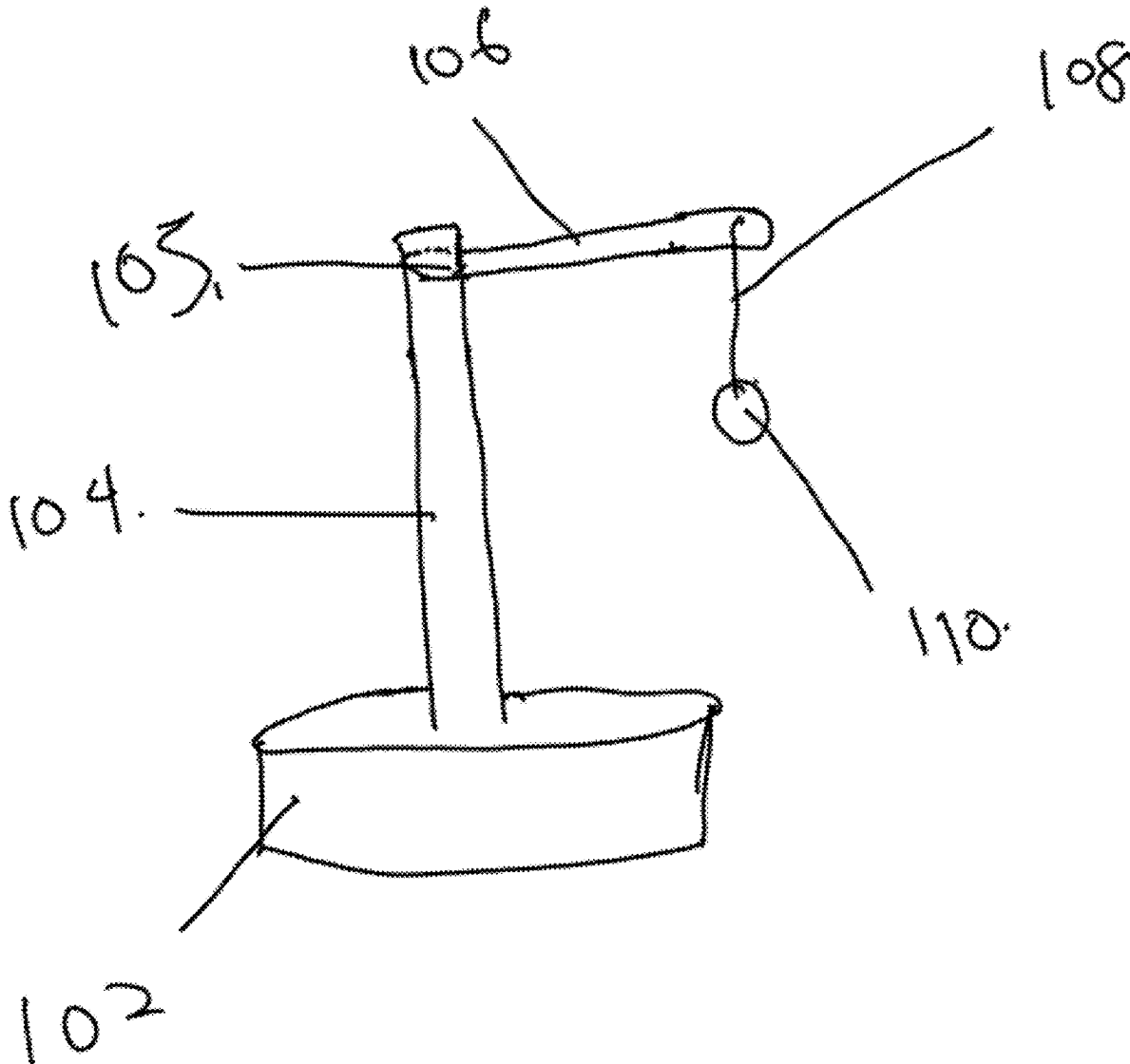
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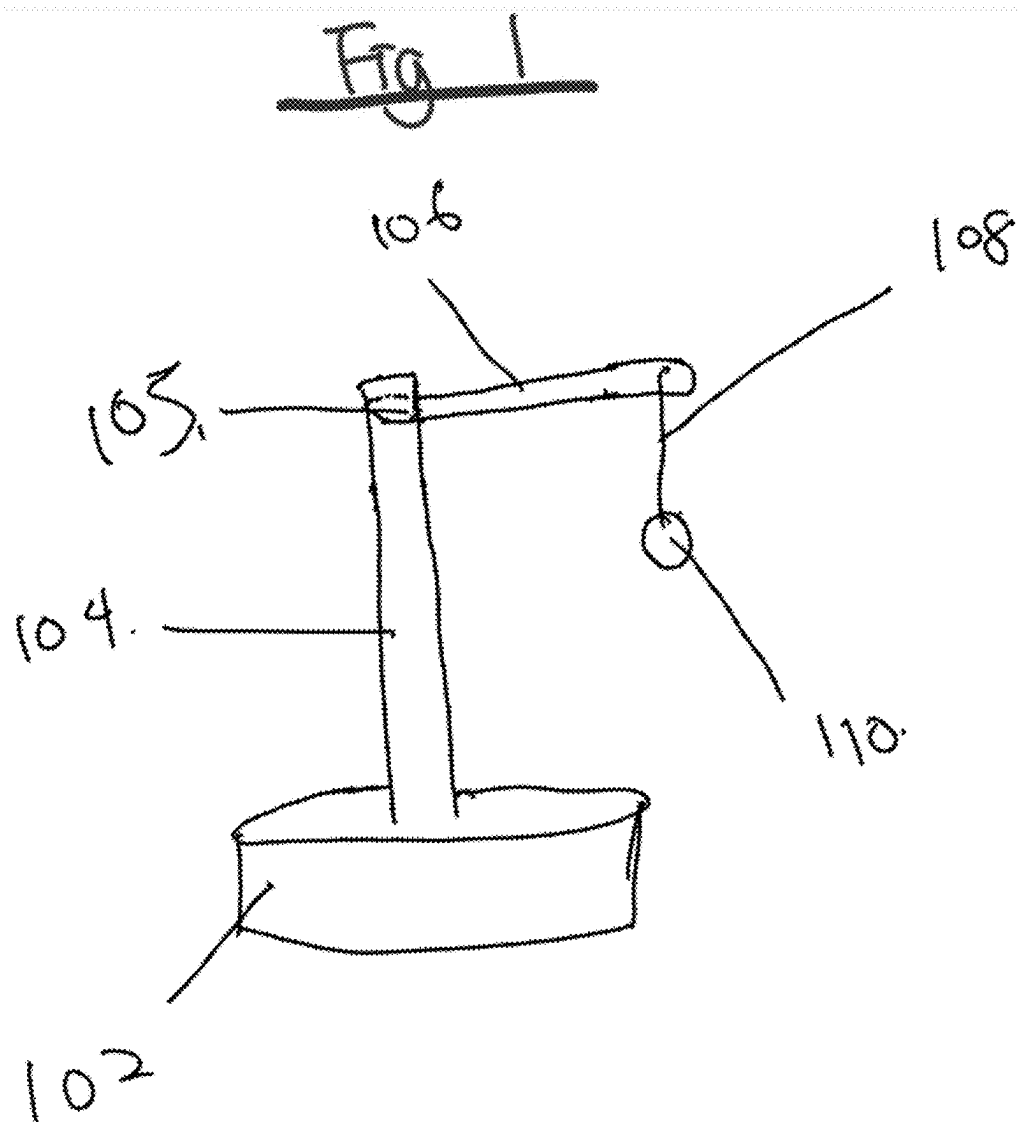
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The apparatus for training volleys comprises of a stable base to keep the apparatus sturdy during use. Connected to the base is a shaft. An arm is connected to the shaft to spin around a substantially vertical axis. A flexible connector is connected to the arm, and a ball is connected to the flexible connector.





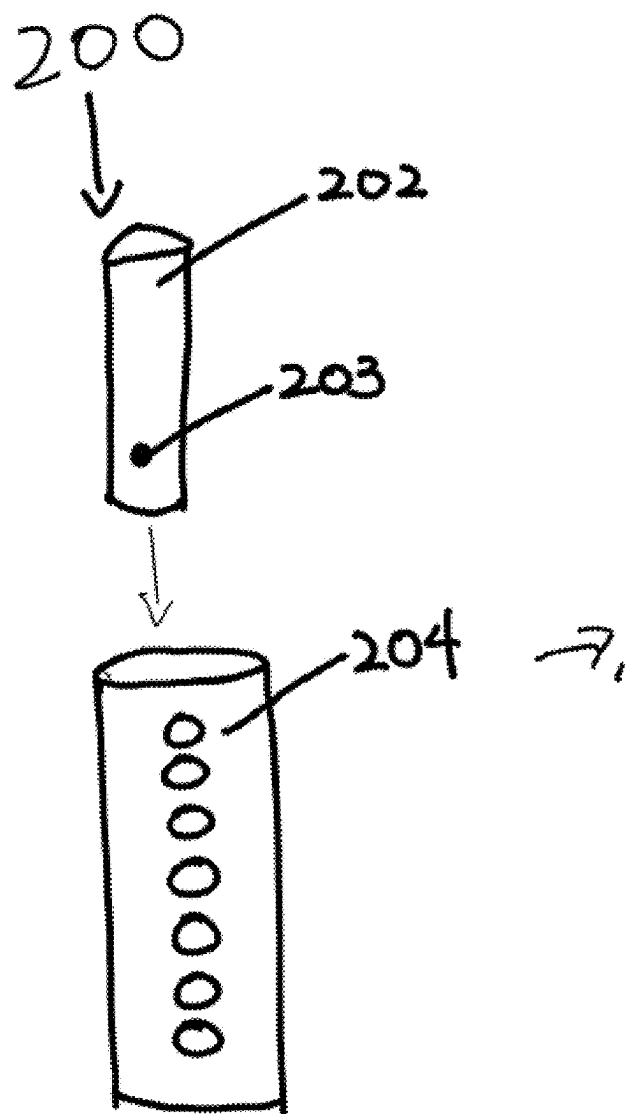


Fig. 2A

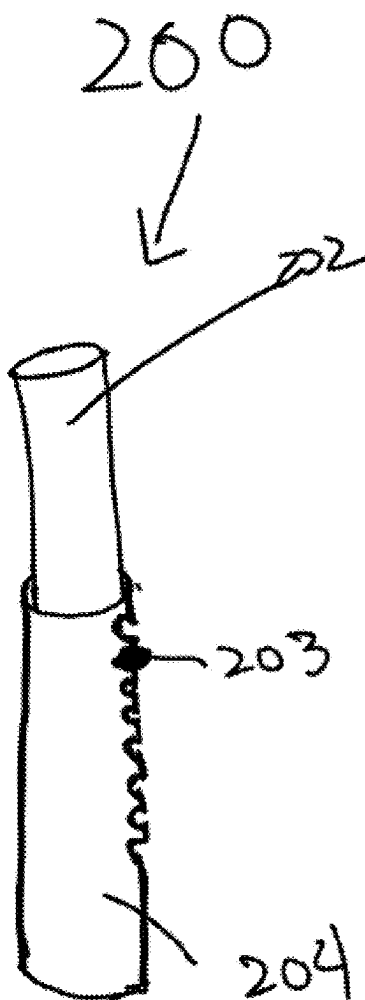


Fig. 2B

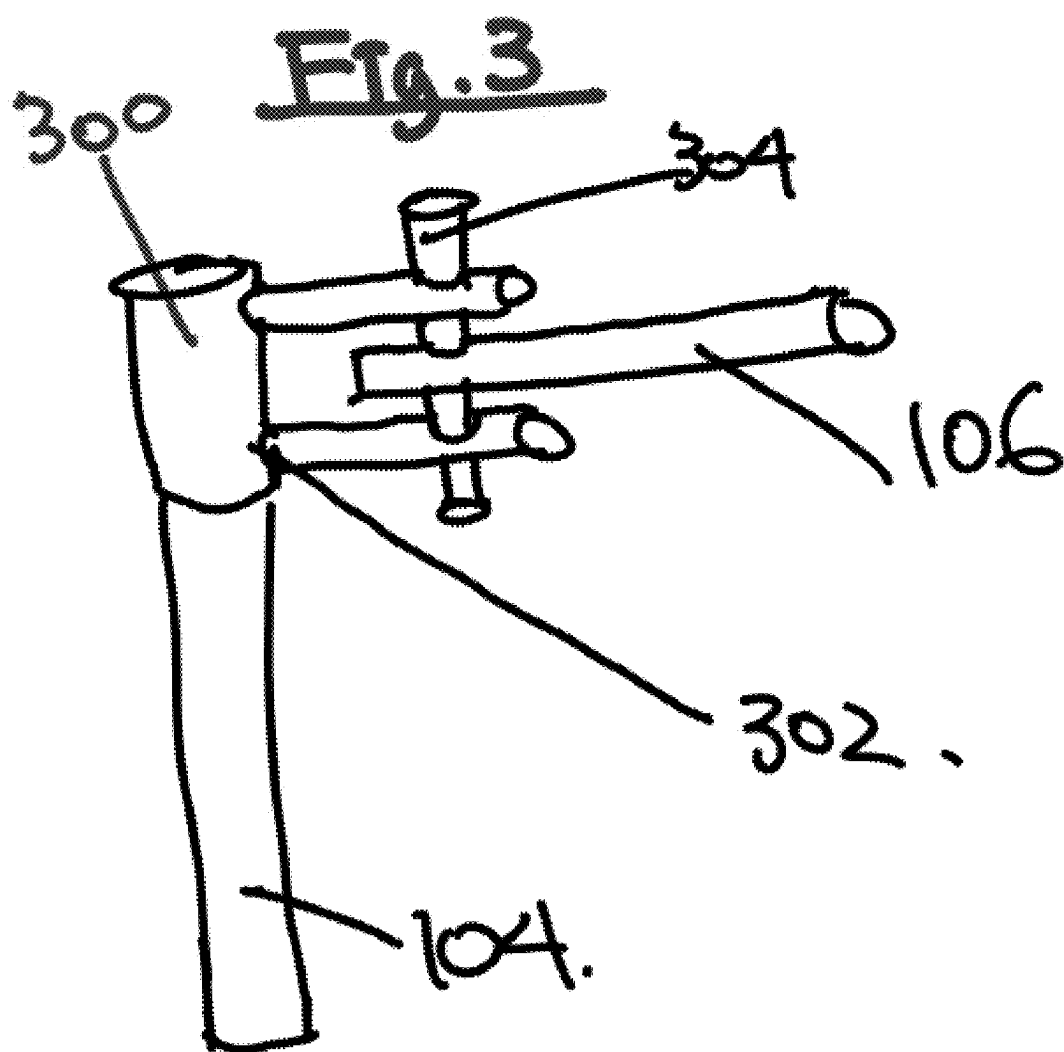
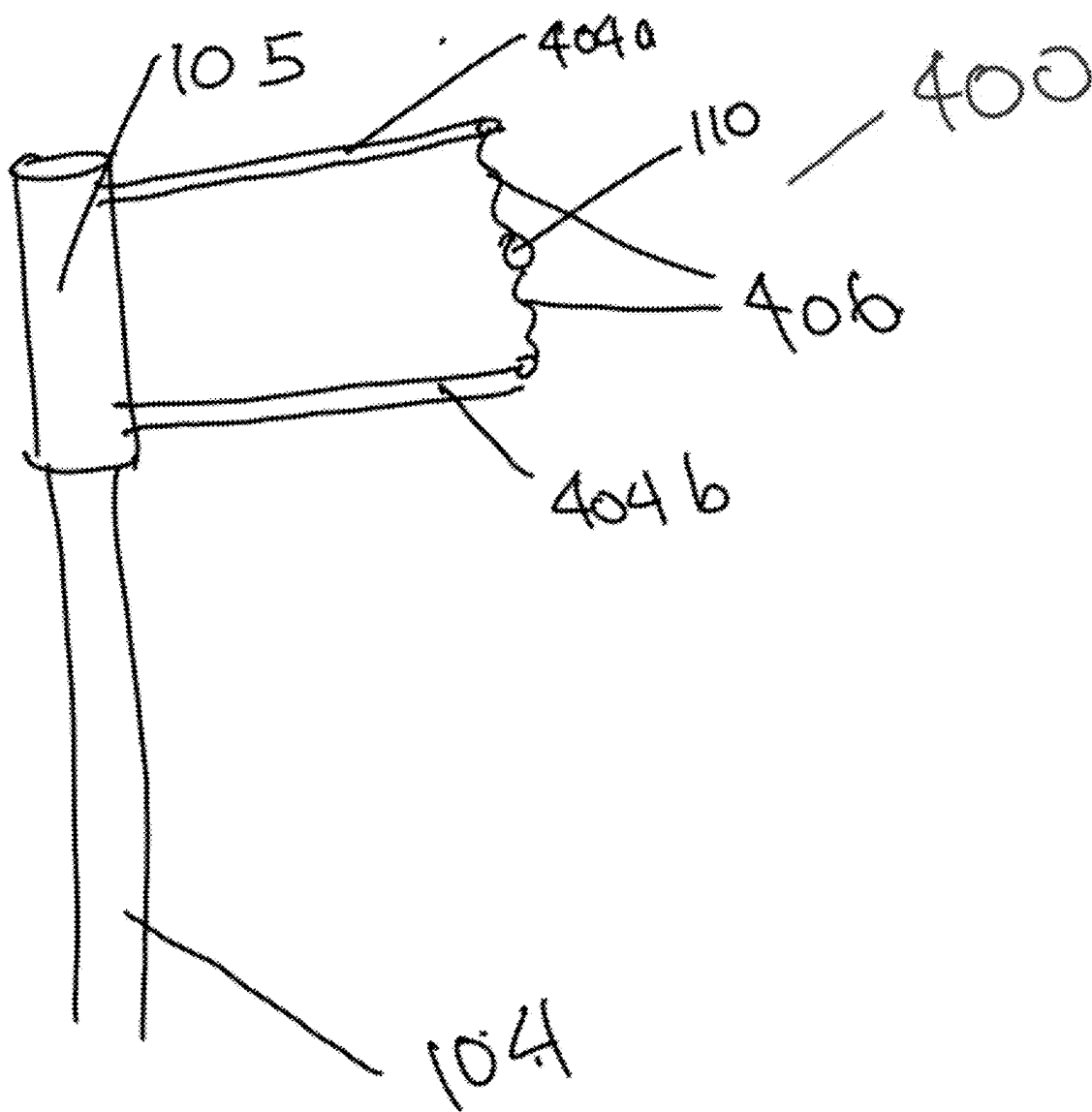


Fig. 4



VOLLEY TRAINING APPARATUS

BACKGROUND

[0001] Tennis is a sport played between two or four players competing on a rectangular court divided by a net. Players use a racket to hit a tennis ball back and forth over the net bouncing no more than once on each side of the net. Tennis consists of many shots, including the serve, the groundstroke, and most importantly, the volley. Whether the surface is clay, grass, or concrete, the players try their best to hit the ball once more than the opponent.

[0002] There have been devices to train different aspects of tennis, such as the Topspin Pro. Topspin Pro is a stationary groundstroke training aid with a free-spinning ball that teaches rhythm, focus, and topspin. Topspin Pro is used by swinging the racket through the ball in an upward motion until the ball produces spin. However, volleys are one of the most important shots in tennis, and the Topspin Pro is ill-suited for practicing volleys.

[0003] Topspin Pro sets the ball low, fit for hitting topspin, but the apparatus cannot imitate real volleys. The Topspin Pro's ball also has a very fast return time which does not give the user enough time to hit the volley, not simulating a proper circumstance for training volleys. Thus, a volley training apparatus is needed to help improve the user's volley skills.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] In the drawings:

[0005] FIG. 1 illustrates an apparatus for tennis training in one embodiment of the invention.

[0006] FIGS. 2A and 2B illustrate an exemplary shaft in the apparatus of FIG. 1 in one embodiment of the invention.

[0007] FIG. 3 illustrates an exemplary arm in the apparatus of FIG. 1 in one embodiment of the invention.

[0008] FIG. 4 illustrates another exemplary arm in the apparatus of FIG. 1 in another embodiment of the invention.

[0009] The use of the same reference numbers in different figures indicates similar or identical elements.

DETAILED DESCRIPTION

[0010] FIG. 1 illustrates an apparatus 100 for tennis training in one embodiment of the invention. Apparatus 100 may be used for practicing various tennis strokes including the forehand volleys and backhand volleys. Apparatus 100 includes a base 102, a shaft 104, a rotary return mechanism 105, an arm 106, a flexible connector 108, and a ball 110.

[0011] Base 102 is sufficiently heavy to keep apparatus 100 stationary when a user strikes ball 110 with a tennis racket. Base 102 may be at least partially solid and made of material of sufficient weight, or it may be hollow to be filled with sand or water of sufficient weight. Base 102 may have a disc or conical shape. Base 102 defines a hole 112 for receiving shaft 104.

[0012] Shaft 104 is secured to base 102. For example, shaft 104's lower end is inserted into hole 112 of base 102. Shaft 104 may be secured frictionally or by a locking mechanism, e.g., a screw, from the bottom of base 102. Shaft 104 may be oriented vertically or even angled away from base 102. Shaft 104 may be straight or curved. Shaft 104 may be solid or hollow. Shaft 104 may have a round or rectangular cross-section. Shaft 104 may be extendable, e.g., between 100 and 200 centimeters, so its upper end is about

the height of a typical volley at the net. The typical height of a net is 92 centimeters, so a height of 100 centimeters is accurate as the ball will not be the same as the net. A height of 180 centimeters is the ideal height as the average height for groundstrokes is 90 centimeters above the net or 180 centimeters above the ground. Shaft 104 may be mechanized or motorized to move its upper end up and down to create different volley heights for the user.

[0013] Rotary return mechanism 105 rotatably couples arm 106's proximal end to shaft 104's upper end so arm 106 spins about an axis that is vertical or substantially vertical, e.g., within 20 degrees of vertical. Rotary return mechanism 105 may be a bidirectional rotary return mechanism that allows arm 106 to rotate both clockwise and counterclockwise from a neutral position and return to the neutral position. Rotary return mechanism 105 may be mechanized or motorized to return arm 106 to the neutral position. For example, bidirectional rotary return mechanism 105 may be a rotary joint with a torsion spring or an elastic band for returning to the neutral position, or a motorized rotary actuator controlled by circuitry to rotate and return to the neutral position. Rotary return mechanism 105 may be secured frictionally or by a locking mechanism, e.g., a screw, to shaft 104 and arm 106.

[0014] Arm 106 may be straight or curved. Arm 106 may be solid or hollow. Arm 106 may have a round or rectangular cross-section. Arm 106 may be jointed. Arm 106 may be extendable, e.g., between 10 to 100 centimeters, to create a variety of volleys.

[0015] Flexible connector 108's upper end is attached to arm 106's distal end, and ball 110 is connected to flexible connector 108's lower end. Flexible connector 108 may be a rope or an elastic band. When ball 110 is struck, it may go in any direction the volleyer has hit and return to the original position with pace, but sufficient travel time, due to the swinging actions of both flexible connector 108 and arm 106, for the volleyer to hit another volley and repeat the process.

[0016] To practice tennis, a user stands beside apparatus 100. The user would hold the racket at a 45-degree angle near the user's ears. By moving the racket forward with pace, the user would strike the ball at an angle to produce a downward spin. As described above, the ball would travel away from the user and then return with pace but sufficient time for the user to another volley and repeat the process.

[0017] FIG. 2A and 2B illustrate a telescoping shaft 200 that may be used as shaft 104 (FIG. 1) in one embodiment of the invention. Shaft 200 includes an extendible part 202 and base part 204. Base part 204 has holes spaced out evenly along its length. Extendible part 202 includes a depressible button that would protrude from one of the holes of base part 204 to lock extendible part 202 to base part 204 at a desired height.

[0018] FIG. 3 illustrates a rotary return mechanism 300 that may be used as rotary return mechanism 105 (FIG. 1) in one embodiment of the invention. Rotary return mechanism 300 includes U-shaped bracket 302 that is connected vertically to shaft 104. Arm 106 is placed in the middle of U-shaped bracket 302 and a pin 304 is inserted vertically through holes in U-shaped bracket 302 and arm 106. This allows arm 106 to give ball 110 (FIG. 1) an articulated motion and to simulate a different volley path.

[0019] FIG. 4 illustrates a double arm mechanism 400 that may be used as arm 106 (FIG. 1) in one embodiment of the

invention. Double arm mechanism **400** includes an upper arm **404a**, a lower arm **404b**, and a flexible connector **406**. Arms **404a** and **404b** are connected to rotary return mechanism **105** with even spacing between the arms. Flexible connector **406** is attached at the opposite ends of arms **404a** and **404b**, with ball **101** fixed in the middle. At the opposite end, flexible connector **406** connects upper arm **404a** to lower arm **404b**, consisting of ball **110** in the middle connected by flexible connector **406**. The mechanism allows for more variation of shots as the ball **110** may bounce back up or down.

[0020] Various other adaptations and combinations of features of the examples disclosed are within the scope of the invention.

What is claimed is:

1. An apparatus for training volleys, comprising:
a base;
a shaft connected to the base;

an arm rotatably connected to the shaft to spin around a substantially vertical axis;

a flexible connector connected to the arm; and
a ball connected to the flexible connector.

2. The device of claim 1, wherein the shaft is vertical.

3. The device of claim 1, wherein the flexible connector is an elastic band.

4. The device of claim 1, wherein the arm consists of two parts, the outside and inside where the outside would have holes and the inside would have a part sticking out, making the length adjustable.

5. The device of claim 1, wherein the ball is attached through a hole in the middle.

6. The device of claim 1, wherein the ball is a tennis ball or a baseball.

7. The device of claim 1, wherein the flexible connector is a rope.

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